

Phase 3 Hybrid Innovation for Predictive Sales: A Reproducible ML-DL Fusion Pipeline with Ablation, Explainability, and Error Segmentation

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Predictive Sales Project

Abstract—This report presents Phase 3 of the Predictive Sales project, where a hybrid model combines classical machine learning and deep learning for B2B sales outcome prediction. The pipeline is fully reproducible through a single command (`python main.py`) and generates analysis artifacts covering model comparison, ablation study, architecture visualization, threshold optimization, SHAP explainability, and business-level error segmentation. The final system fuses probabilities from Logistic Regression and a custom DualPathMLP using weighted averaging (0.4 and 0.6). Experimental evidence confirms that both model components contribute complementary strengths: the deep model supports high recall, while the classical model helps moderate false positives in the fused output.

Index Terms—B2B sales prediction, hybrid learning, dual-path MLP, ablation study, SHAP, threshold optimization, error segmentation

I. INTRODUCTION

Predicting sales opportunity outcomes is a high-impact task for pipeline prioritization and resource allocation. In practical sales operations, model value is determined not only by global metrics but also by decision quality under business constraints. Phase 3 addresses this by introducing a reproducible **hybrid innovation pipeline** that integrates:

- an interpretable machine learning branch (Logistic Regression),
- a non-linear deep learning branch (DualPathMLP), and
- decision-centric analyses beyond aggregate score reporting.

II. PHASE 3 OBJECTIVES

The main objectives were:

- 1) implement a hybrid model using weighted probability fusion,
- 2) validate component contribution through ablation,
- 3) generate a clear architecture diagram,
- 4) provide explainability via SHAP,
- 5) optimize decision threshold,
- 6) segment model errors by business dimensions,
- 7) ensure single-command reproducibility.

III. DATA AND FEATURE SETUP

The model uses processed tabular CRM data from:

`data/processed/processed_sales_data.csv`.

(1)

The feature set contains 14 variables split into:

- **Numeric (8):** year_established, revenue, employees, sales_price, engage_year, engage_month, log_revenue, log_employees.
- **Categorical (6):** sector, office_location, series, product, manager, regional_office.

A stratified 80/20 train-test split with random state 42 is used throughout.

IV. METHODOLOGY

A. ML Branch

The classical branch is a pipeline with:

- `StandardScaler` for numeric features,
- `OneHotEncoder(handle_unknown='ignore')` for categorical features,
- `LogisticRegression(max_iter=2000, class_weight='balanced')`.

B. DL Branch: DualPathMLP

DualPathMLP learns separate representations for numeric and categorical signals, then fuses them:

$$\mathbf{h}_{num} = \text{ReLU}(W_{num}\mathbf{x}_{num} + b_{num}), \quad (2)$$

$$\mathbf{h}_{cat} = \text{ReLU}(W_{cat}\mathbf{x}_{cat} + b_{cat}), \quad (3)$$

$$\mathbf{h}_{fuse} = [\mathbf{h}_{num}; \mathbf{h}_{cat}], \quad (4)$$

$$\hat{y}_{dl} = \sigma(f(\mathbf{h}_{fuse})). \quad (5)$$

Training uses binary cross-entropy with L2 penalty, Adam optimization, mini-batches, and validation-based early stopping.

C. Hybrid Fusion

Final probability is computed as:

$$\hat{y}_{hybrid} = 0.4 \hat{y}_{ml} + 0.6 \hat{y}_{dl}. \quad (6)$$

Default decision threshold is 0.5 for baseline comparison, followed by threshold sweep analysis.

V. EXPERIMENTAL RESULTS

A. Model Comparison at Threshold 0.5

TABLE I
ML VS DL VS HYBRID PERFORMANCE

Setup	F1	Precision	Recall	ROC-AUC
DL only	0.7660	0.6317	0.9729	0.5206
Hybrid	0.7450	0.6276	0.9163	0.5196
ML only	0.4728	0.6430	0.3738	0.5140

Additional confusion-level view:

- DL only: FP=481, FN=23
- Hybrid: FP=461, FN=71
- ML only: FP=176, FN=531

B. Ablation Findings

Ablation confirms complementary contributions:

- Removing DL (ML only) reduces recall by 59.20% relative to hybrid.
- Removing ML (DL only) increases false positives by 4.34% relative to hybrid.

Thus, hybrid design improves decision balance rather than maximizing a single metric.

VI. THRESHOLD OPTIMIZATION

Hybrid thresholds were evaluated at 0.3, 0.4, 0.5, 0.6, and 0.7.

TABLE II
HYBRID THRESHOLD SWEEP

Threshold	F1	Precision	Recall	ROC-AUC
0.3	0.7741	0.6314	1.0000	0.5196
0.4	0.7741	0.6314	1.0000	0.5196
0.5	0.7450	0.6276	0.9163	0.5196
0.6	0.3933	0.6842	0.2759	0.5196
0.7	0.0949	0.7414	0.0507	0.5196

The best F1 occurs at threshold 0.3, showing that the business objective can shift the operating point toward recall-intensive strategy.

VII. EXPLAINABILITY VIA SHAP

SHAP analysis on the ML branch produced top global features:

- 1) num_log_employees (0.0950)
- 2) num_engage_year (0.0800)
- 3) num_year_established (0.0751)
- 4) num_log_revenue (0.0574)
- 5) num_revenue (0.0353)

Categorical signals among top features include office_location_Panama and regional_office_Central, indicating meaningful regional effects in win propensity.

VIII. BUSINESS ERROR SEGMENTATION

Misclassification rates were segmented by sector, product type, and price range. Critical observations:

- Highest segment error appears in Price Range (1096, 3393] at 46.79%.
- Product MG Advanced also shows 46.79% error.
- Services sector shows 45.57% error.

This analysis translates model outputs into direct targeting, qualification, and go-to-market improvement opportunities.

IX. REPRODUCIBILITY AND ARTIFACTS

Phase 3 is executed by:

python main.py (7)

The pipeline automatically generates all artifacts in results/, including:

- hybrid_model_comparison.csv
- ablation_study_metrics.csv, ablation_study_grouped_bar_chart.png
- threshold_analysis.csv, threshold_analysis_hybrid.png
- shap_summary_top10_ml.png, shap_top10_features_ml.csv
- error_segmentation_summary.csv, error_segmentation_by_category.png
- architecture_diagram.png

X. CONCLUSION

Phase 3 successfully delivers a robust and reproducible hybrid innovation layer on top of earlier work. The main contribution is not only model fusion, but also comprehensive decision support: ablation-backed model design, threshold-driven operating policy, SHAP explainability, and business-segmented error diagnostics. The resulting system is suitable for both academic reporting and practical sales analytics deployment.

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